

```

clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

```

Filter Order = 6

```

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

```

Cutoff Frequency = 0.40 * pi rad/sample

```

if all(abs(roots(a))<1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

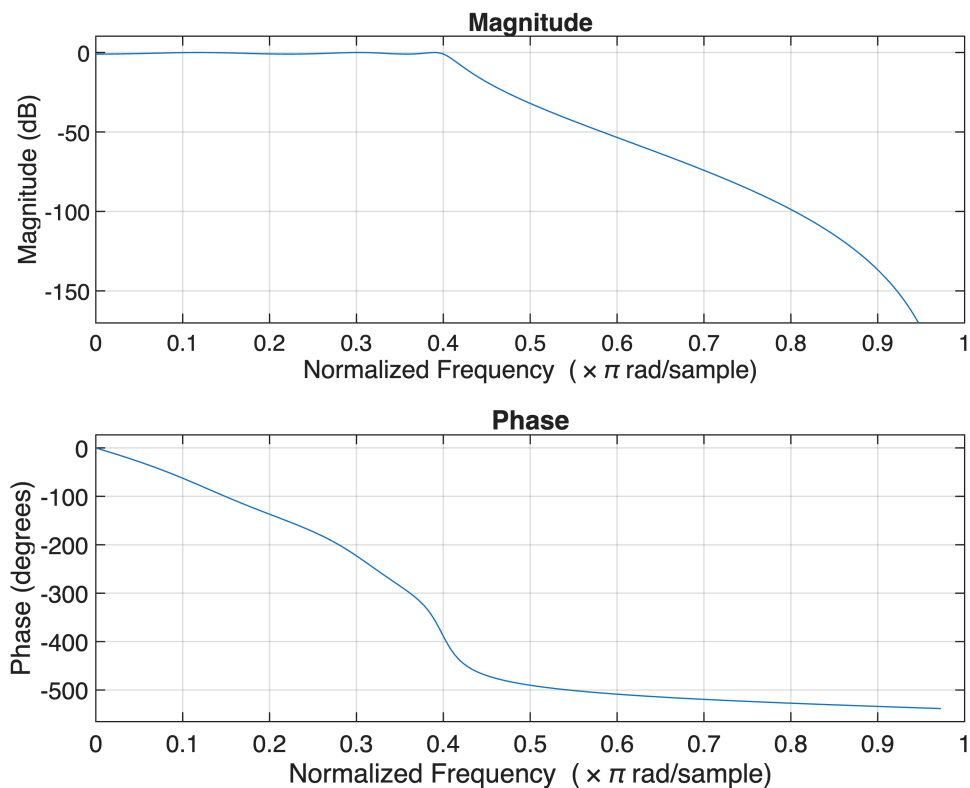
```

System is STABLE

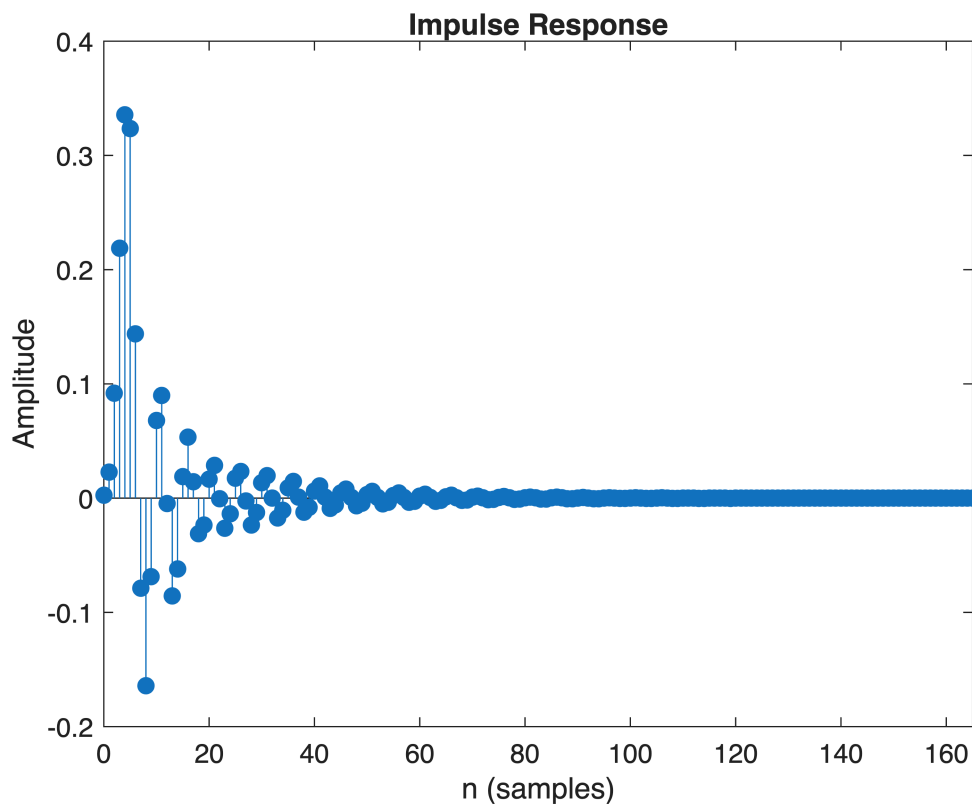
```

figure; freqz(b,a);

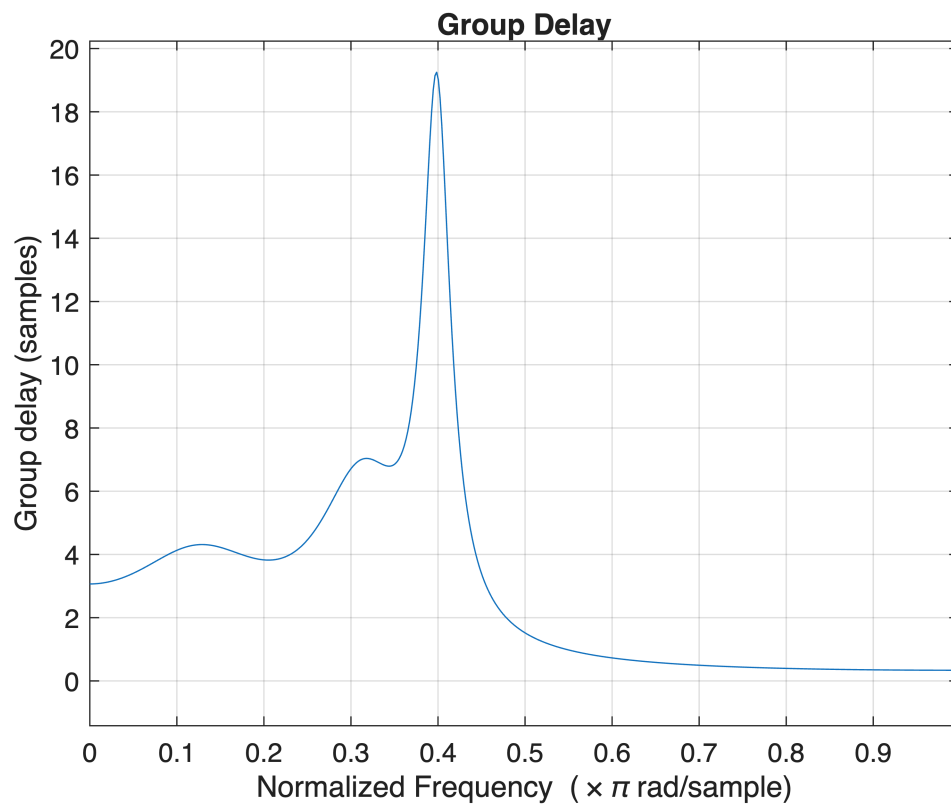
```



```
figure; impz(b,a);
```



```
figure; grpdelay(b,a);
```



```
figure; zplane(b,a);
```

